

AI FitTrack Project Planning

Phase 4

This plan organizes the completed backend work into dependency-based gates and records the confirmed four-person team structure, academic details, and confirmed responsibilities.

Field	Value
Project	AI FitTrack
Submission	SmartBridge Naan Mudhalvan
Team ID	SWTID-2026-8100
Team size	4
College	DRBCCC Hindu College
Department	B.Sc. Computer Science with Artificial Intelligence
Academic Year	2024–2027
Faculty Guide	Prof. D Saraswathy
Submission Date	
Team Leader	Parthiban M (NM ID: 28538B1DB05060E8D804CDAA88169909, Reg: 222401226)
Team Members	Naveen J (NM ID: 53C2DC1753C6DFDAF8649841B4630C80, Reg: 222401223); Nidesh N S (NM ID: FDE55A8DA9135F6AB7347C01AD1B8A17, Reg: 222401224); Nithish Kumar P (NM ID: BFBD9AD8CEC72229F374877DB82CC0B8, Reg: 222401225)
Project scope	Backend-only REST API
Technology	Node.js, Express.js, MongoDB, Mongoose, JWT and Google Gemini AI
Source baseline	Latest uploaded backend ZIP and supplied verification evidence
Document status	Submission-ready verified technical documentation with complete academic details

Prepared for academic evaluation. Secrets, credentials, private environment files and unverified claims are excluded.

Document contents

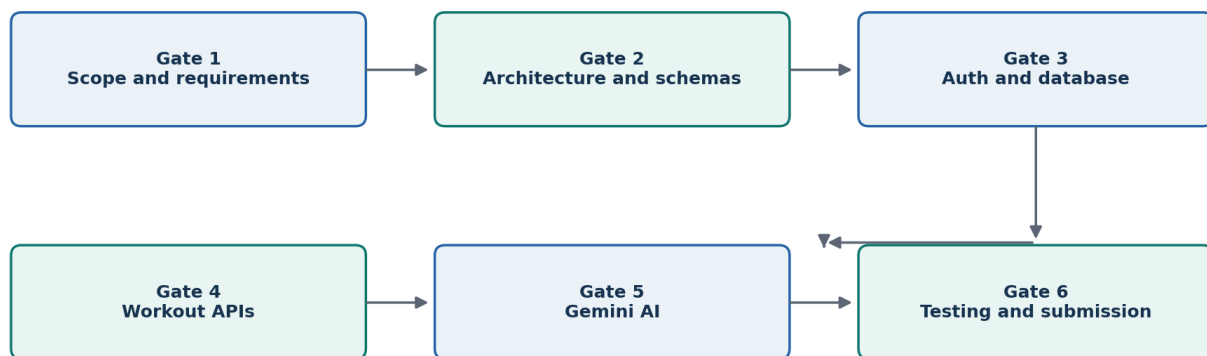
1. Planning approach
2. Work breakdown
3. Deliverables
4. Dependencies
5. Quality and risk plan
6. Roles and submission gates

Planning approach

This plan organizes the project using ordered completion gates with measurable outputs and acceptance conditions. The confirmed team roster, academic details, and confirmed responsibilities are recorded below.

Evidence-based project plan

Sequence uses dependency gates instead of unverified calendar dates



Actual dates and assignees are intentionally left for the student to confirm

Figure 1 Dependency-based project gates without unverified dates

Work breakdown structure

Gate	Work package	Deliverable	Acceptance evidence
1	Scope and requirements	Problem statement, objectives, actor and requirement list	Official document mapped to 24 backend requirements
2	Architecture and data	Layer design, API design, User and Workout schemas	Source folders, schema constraints and diagrams agree
3	Authentication and database	MongoDB connection, User model, bcrypt, JWT, profile	Registration, login, profile and unauthorized evidence

Gate	Work package	Deliverable	Acceptance evidence
4	Workout module	CRUD, search, validation and ownership	Eight live workout records plus automated tests
5	Gemini AI module	Recommendation, submitted insights and database insights	Three live sanitized Gemini response records
6	Verification and submission	Tests, documentation, repository and demonstration	95 automated passes recorded; final links still pending

Planned deliverables

- Backend source with dependency lockfile and safe .env.example.
- Postman collection containing health, authentication, workout and AI requests.
- Automated tests for authentication, workouts, validation, security and Gemini retry behavior.
- Sanitized live verification records with no passwords, tokens, database credentials or API keys.
- Eight-phase DOCX and PDF documentation set.
- Public GitHub repository and student-recorded demonstration video after final upload.

Dependency plan

Dependency	Needed by	Control
Node.js 20+ and npm	All development and tests	Use engines declaration and package-lock.json.
MongoDB	Auth and workout persistence	Connect before server listen; fail with actionable error.
JWT secret	Registration, login and protected routes	Fail at startup if absent; recommend at least 32 characters.
Gemini API key	AI recommendation and insights	Return safe 503 when absent; do not expose the key.
Network access	Atlas and Gemini live checks	Use local mocked tests when external services are unavailable.
Postman or HTTP client	Demonstration and manual verification	Run the collection in the documented order.

Risk register

Risk	Likelihood	Impact	Implemented or planned response
Credential exposure	Medium	High	Commit only .env.example; hide terminal and Postman environment values during recording.

Risk	Likelihood	Impact	Implemented or planned response
Atlas connection failure	Medium	High	Configurable DNS, IPv4 option, timeout and clear startup failure.
Gemini quota or downtime	Medium	Medium	Bounded retries, timeout, rate limit and safe normalized errors.
Cross-user data access	Low	High	JWT user reload and owner-scoped queries; automated isolation tests.
Invalid client input	Medium	Medium	Controller validation plus Mongoose validation and standard 400 responses.
Evidence mismatch after code change	Medium	Medium	Re-run Postman evidence after the final ZIP is frozen.
Missing submission identity or URLs	High	Medium	Complete the pending metadata checklist before upload.

Quality plan

- Review requirement-to-source traceability before declaring completion.
- Keep password, token and API-key values out of evidence and documentation.
- Run automated tests, smoke checks, Atlas checks and live Gemini checks on the frozen release.
- Verify response envelopes and status codes in Postman.
- Inspect DOCX and PDF output for clipping, unreadable tables and missing figures.
- Confirm the public repository excludes node_modules and every private .env file.

Team roster and responsibility status

Team member	Supplied designation	Confirmed responsibility
Parthiban M	Team Leader	Project architecture, complete backend development, MongoDB integration, Gemini AI implementation and final delivery.
Naveen J	Team Member	Reviewing requirements, organizing the eight-phase documentation and checking submission guidelines.
Nidesh N S	Team Member	Reviewing Postman API test results, checking test evidence and helping verify that the APIs meet the requirements.
Nithish Kumar P	Team Member	Reviewing the final report, helping prepare the demonstration and organizing GitHub and SkillWallet submission materials.

The confirmed responsibilities of each team member are listed in the team roster below.

Items requiring student confirmation

Pending item	What to provide
Repository	Insert the final public GitHub repository URL after upload.
Deployment	Insert the deployed API base URL if the evaluator requires a hosted demonstration.
Demonstration video	Record the video using Phase 8 and insert the final shareable video URL.